

Retroflex

Digital rideshare identification

A stationary-first display, controlled by the driver.

An Android app paired with an in-vehicle Android terminal.
Working prototype; demonstration available.

SEEKING A CO-DESIGN SPONSOR

Independent concept. No TfNSW or rideshare-platform endorsement is claimed.



A practical opportunity—if it can be proven safe

Explore a digital companion to existing rideshare identification, not a substitute for it.

01

TODAY

Physical identifiers are fixed

Rideshare identification is commonly provided by a physical sticker or other fixed identifier.

02

QUESTION TO TEST

Can the display follow vehicle state?

Could an authorized identifier be driver-managed and automatically go dark when movement begins?

03

PILOT OPPORTUNITY

Start with evidence

A supplementary display can be assessed against road safety, clarity, usability, privacy, and cyber requirements.

PROPOSED SAFEGUARD

Keep every required physical sticker or identifier in place unless formally approved otherwise.

One Android app. One paired in-vehicle terminal.

A simple operating model for driver control, display status, and a customer-facing identifier.

01

DRIVER APP

Android mobile

Enable or disable the display, choose an authorized identifier, check terminal power/charging, and set brightness.

02

REMOTE CONTROL

Internet pairing

Commands are sent to the paired terminal. Authentication, command integrity, and outage handling require independent review.

03

DISPLAY TERMINAL

Android mobile or tablet

Shows the approved identifier. Mounting position(s) are agreed only after visibility, distraction, and occupant-safety assessment.



PILOT PRINCIPLE

Retain required physical rideshare identification during assessment and any approved pilot.

Display state follows vehicle state

Proposed behavior for validation; the speed source, response time, and fault handling must be tested.



These parameters are prototype/proposed requirements—not a finding of NSW legal compliance.

A pilot should measure benefits—not assume them

The public-value case is a hypothesis to test with drivers, passengers, and independent reviewers.

01

PICKUP CLARITY

Can participants identify the intended rideshare vehicle accurately—and how long does it take?

02

MOTION-STATE RELIABILITY

Does the screen blank promptly when movement begins and return only after the agreed stop period?

03

DRIVER EXPERIENCE

Is the app practical before a trip, without causing distraction, glare, or added workload?

04

DEVICE RELIABILITY

How well do pairing, charging, brightness, connectivity, and fault handling work in real conditions?

No safety, efficiency, or customer-experience benefit should be claimed until independently evaluated.

Why sponsor the feasibility stage?

A small, conditional investment can generate decision-quality evidence before any larger commitment.

01

EVIDENCE BEFORE DEPLOYMENT

Independently assess road safety, vehicle fit, privacy, usability, and cybersecurity before any on-road trial.

02

PUBLIC-INTEREST LEARNING

Measure pickup clarity, motion-state behavior, and driver experience using jointly agreed criteria.

03

RISK REDUCTION

Identify legal, safety, technical, and cyber issues early; define mitigations and an appropriate pathway.

04

NO LOCK-IN

Feasibility sponsorship does not commit government to procure, endorse, or roll out the product; it creates a clear go/no-go decision.

Sponsorship funds learning and risk reduction; any deployment remains evidence-led and separately approved.

Safety, privacy, and cyber review come first

The prototype is pre-pilot. Independent assessment and written approvals come before any road deployment.

ROAD & VEHICLE SAFETY

01

- Confirm the road-rule, vehicle-standard, and signage pathway before any public-road use.
- Assess field of view, distraction, glare/night visibility, mounting, wiring, and occupant safety.
- Validate blanking on motion and on sensor, connection, or system faults.
- Phase one: authorized ride identifiers only—no ads, QR, video, animation, or audio.

CYBERSECURITY & PRIVACY

02

- Test device pairing, authentication, access control, command integrity, and data protection.
- Review backend access, software updates, logging, resilience, and outage behavior.
- Threat-model and penetration-test; close critical findings before a live pilot.
- Minimize data; define consent, retention, privacy, and incident-response controls.

Proposed controls—not evidence of compliance or certification. Effectiveness must be independently verified.

Three gates from demonstrator to evidence

Proceed only with written approvals, agreed measures, and a clear stop/rollback process.

01

CO-DESIGN & FEASIBILITY

- Name a sponsor and working group.
- Map rules, dependencies, stakeholders, and risks.
- Agree independent reviewers and decision criteria.

02

BENCH & OFF-ROAD VERIFICATION

- Prove speed threshold, blanking, reactivation, and fault behavior.
- Test brightness, power, mounting, and device compatibility.
- Complete privacy and cybersecurity assessment.

03

CONTROLLED, OPT-IN EVALUATION

- Proceed only under written approvals and agreed conditions.
- Keep required physical identifiers in place.
- Monitor outcomes; report openly and retain rollback ability.

Pilot size, duration, and location are decided jointly after the feasibility and verification gates.

A co-work model with independent oversight

A practical arrangement that shares the evidence and keeps decision rights clear.

WHAT WE'RE ASKING TfNSW TO CONSIDER

- Nominate a sponsor or appropriate program contact.
- Connect relevant safety, vehicle-standards, digital, privacy, and cyber teams.
- Co-design requirements, approval pathway, measures, and pilot conditions.
- Consider support for independent assessment if the feasibility case proceeds.
- Advise whether an eligible innovation grant, prize, or project-originator award is available.

WHAT THE PROJECT TEAM CAN CONTRIBUTE

- Provide the working demonstrator and live walkthrough.
- Supply product requirements, Android build, test evidence, and engineering support.
- Iterate against approved constraints and independent recommendations.
- Share the risk register, findings, and go/no-go results transparently.

Any grant, prize, bounty, or remuneration is subject to TfNSW/relevant authority project approval, eligibility, evaluation, and written terms—not an entitlement.

Let's agree on the first deliverable.

A joint feasibility plan—not a promise to deploy.

01 DEMONSTRATE

45-minute prototype briefing with the relevant TfNSW teams.

02 DEFINE

Confirm rules, independent assessors, risks, and conditions for testing.

03 DECIDE

Agree measures, governance, funding needs, and written go/no-go gates.

Would Transport for NSW be open to a short scoping meeting?

Working prototype and demonstration available now.

[Your name] · [Email] · [Phone]

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